	FDG model (1)	FJ model (4)
Resistance to the others	self-weight a_{ii}: inertia, a weight on the current opinion	attachment $1-\lambda_i$ to the prejudice u_i: an anchor to a fixed value
Stubborn agent (Definition 2.4)	$a_{ii}=1$: a condition on the matrix only	$\lambda_i=0$ and $u_i=x_i(0)$: a condition on the matrix and on the input
Opinion of a stubborn agent	a state	an input
Convergence	Table 1: sinks of $C(G)$ and their aperiodicity	all agents P-dependent (Corollary 3.3), or W^{22} regular (Theorem 3.2)
Asymptotic stability	never: the eigenvalue 1 lies on the unit circle (Theorem 1.7)	if and only if all agents are P-dependent, i.e. $\rho(\Lambda W)<1$ (Corollary 3.3)
Final opinions	consensus $(w^\top x(0))\mathbf{1}_n$ (Theorem 2.1), or disagreement with at least two aperiodic sinks (Theorem 2.3)	if all agents are P-dependent, $x(\infty)=Vu$ (9): convex combinations of the prejudices, independent of $x(0)$
Strongly connected network	consensus or no convergence, never disagreement	disagreement whenever the prejudices that count differ
Aperiodicity	always needed	needed only for the P-independent agents
Measure of power	social power w (2): requires a consensus	influence centrality c (12): requires $\rho(\Lambda W)<1$ and $u=x(0)$
From one model to the other	with stubborn agents, an FJ model with $\lambda_i=0$, $u_i=x_i(0)$ for the stubborn agents [5]	an FDG model on the augmented graph (11), with n virtual stubborn agents

Table 3: Comparison between the FDG model and the FJ model.

Inertia and anchor. An agent can resist the opinions of the others in two ways, one for each model. A self-weight a_{ii} is inertia: an agent whose neighbours all hold the opinion 1 updates as $x_i(k+1)=a_{ii}\,x_i(k)+(1-a_{ii})$ and reaches 1, only more slowly as a_{ii} grows. An attachment $1-\lambda_i$ is an anchor: the same agent updates as $x_i(k+1)=\lambda_i+(1-\lambda_i)\,u_i$ and stops at once between 1 and its prejudice, never reaching the others. This holds for a single agent on any network; the effects on the group are shown on the two networks of Figure 1, both strongly connected and periodic without self-loops.

- *Convergence.* In the FDG model the self-weight of agent 1 is needed to make the network aperiodic (Section 2.2); in the FJ model a single prejudiced agent reached by everybody makes all agents P-dependent, and the model converges although the network is periodic (Corollary 3.3).
- *Influence.* The social power of agent 1 grows gradually with its self-weight, from 1/6 to 1 (Figure 3), and for $t<1$ the consensus still depends on the initial opinions of all agents. A single anchor, however loose, takes the whole influence centrality, $c=e_1$, and the group ends on its prejudice (Figure 9).
- *Many agents.* With the same weight 0.2 for every agent (Figure 10), the FDG model reaches a consensus, as on every strongly connected network with a self-loop (Corollary 1.14), while the FJ model keeps a disagreement as soon as the prejudices differ, on any network.

In social terms, inertia buys time and a gradual influence, but in the end the agent follows the group; an anchor buys permanence: the agent never gives in completely, a single anchor takes all the influence, and different anchors keep the group divided.

State and input. The same opinion can be a state or an input, and this decides the stability. An FDG model with stubborn agents and its FJ rewriting [5] have the same trajectories, but the FDG model is never asymptotically stable, while the FJ model is (Corollary 3.3): the opinion of a stubborn agent is a state in the first and the input $u_i = x_i(0)$ in the second. The spectra show it in one step: the stubborn row is $e_i^\top$ in A and zero in ΛW, so the eigenvalue 1 of the FDG model becomes 0 in the FJ model. Conversely, the augmented model (11) turns every prejudice back into the state of a virtual agent, adds one eigenvalue 1 for each of them, and loses the asymptotic stability.

- *What is perturbed.* Moving $x_1(0)$ (Figure 7) moves a state in the FDG model, which keeps it forever, and only the initial opinion of a real agent in the FJ model, which forgets it and returns to the same limit Vu. Moving u_1 would change the FJ limit as well.
- *Memory without stubbornness.* A closed group of P-independent agents is a memory inside the real agents (Figure 12): it keeps its initial opinions as a virtual agent keeps its prejudice, so the FJ model converges without being asymptotically stable.

In social terms, in the FDG model stubbornness is a memory inside the group, which is never forgotten; in the FJ model it is an external influence, and the group forgets its initial opinions: what the group remembers is what the model counts as its state.

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